

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electrical and Electronics Engineering
Academic Year: 2018 - 2019 (Even Semester)

Innovative Practices Description

UNIT III Electrical Machines

Degree, Semester & Branch: II Semester B.E Mechanical 'A'

Course Code & Title: BE8253 Basic Electrical, Electronics and Instrumentation Engineering

Name of the Faculty member: S. Meenakshi Sundaravel

Name of the Topic: Construction of DC Machines

Name of the Innovative Practice: Demonstration of construction of dc machine

Date & Duration: 01.03.2019, 25 minutes

ICT Tool Used: Demonstration using machine model

Description:

For explaining the constructional details of dc machine, it is better to demonstrate with the dismantled dc machine with various hardware components for better understanding.

Goals (Learning Outcomes):

- At the end of this practice, students can be able to list various parts of dc generator/motor clearly.
- At the end of this practice, students can be able to demonstrate the constructional details of dc generator/motor.

Use of appropriate methods:

Justification for choosing the activity for the topic:

The topic "Construction of dc machines" includes various parts and the names of those parts are difficult to remember for mechanical engineering students. To make it easier, i have chosen this activity by showing various hardware parts individually through a dismantled dc machine demo model.

Effective presentation

Details of the Implementation:

Time allotted for this activity – 25 minutes

For this activity, the demo machine has been taken to the classroom for demonstration. The classroom is divided into 2 batches and for the two batches, I have explained the constructional details and function of each components separately.

Total number of students: 38, 1st Batch: 19 students, 2nd Batch: 19 students

Significance of Results:

Assessment of Effectiveness/Success of the Activity:

- The assessment of effectiveness of the activity was felt when the students listed all the parts clearly after this activity.
- The success of the activity was evaluated by asking the same question in Internal Assessment Test II – Around 68.75% of the students answered correctly.

Reflective Critique:

- **Benefits:**

The students can be able to demonstrate various parts of dc generator/motor and the functions of each parts clearly.

- **Challenges:**

Many students are expecting the demo model for explaining the construction of ac machines also.

CO	PO1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3	PSO 4
CO3	3	-	-	-	1	-	-	-	-	-	-	1	-	2	-	-

CO3: Demonstrate the construction and working principle of various electrical machines.

References:

1. Thereja.B.L., - Fundamentals of Electrical Engineering and Electronics, S. Chand & Co. Ltd., 2008.